

Notes

Unit 3 - *Internal influences on memory*

Lecture 9: *Cognitive Control* - “top-down” influences on learning and memory.

Lecture 10: *Forgetting* - How, why, when

Lecture 11: *Distortion and False memory*

Lecture 12 (today): *Emotional memories*

Quiz #3 on Thursday!

Lecture 12: Emotional memories

PSYCH/COGS 140M 2026.05.19

Review think/no-think results

- **No-Think** instruction → Reduced memory for **paired** word (**X-Y**)
- Different from “don’t think of an elephant”
- Instead: “don’t follow the train of thought that leads to elephant.”
- “Don’t think of Y” → Brings Y to mind!
- “Don’t think of what follows X” → *cognitive control inhibits Y*

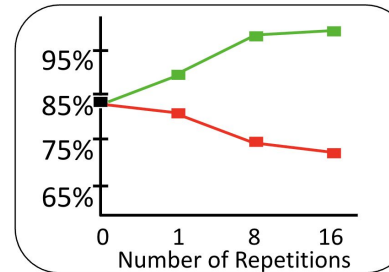
- Big difference!

Think / No-Think Task

STUDY	Ordeal - Roach
	Journey - Pants
	Flag - Sword

THINK /	Flag -
NO-THINK	Ordeal -

TEST	Ordeal - R_
	Journey - P_
	Flag - S_



More practice ‘not remembering’ (No Think trials) leads to worse memory when items are later tested³

Outline: Emotion and Memory

- Defining emotion
- Manipulating and measuring emotion
- Emotion and conditioning
- Emotion and declarative memory

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- Defining emotion
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What is emotion?

- An internal and subjective state that we experience and infer in others
 - Helps us adaptively react to a situation
 - Helps us communicate
- Emotions consist of:
 - **Bodily responses**
 - Physiological changes (e.g., increased heart rate)
 - Movement (e.g., change in facial expression)
 - **Conscious feelings** (e.g., being afraid)

Basic emotions

- Influential work by Paul Ekman suggests that humans share a small set of basic emotions:
 - Happiness, sadness, anger, fear, disgust, surprise



Theories of emotion

- We've defined emotions as having two major components:
 - **Bodily responses** and **conscious feelings**
- How do these components relate to one another?
 - Different theories suggest different relationships

James-Lange theory (1884)

- Conscious emotional feelings occur when we sense a particular combination of bodily responses

(a) James-Lange theory



- i.e., emotions are labels that we assign to the body's arousal

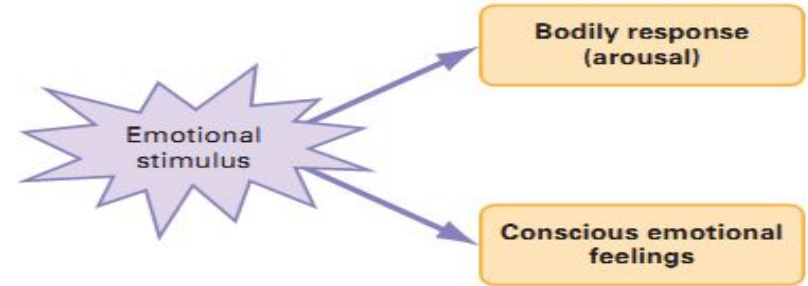
Cannon-Bard theory (1915)

- Objected to the J-L theory for several reasons:
 - Many types of arousal are quite similar across emotions (e.g., racing heart)
 - Experiencing arousal doesn't necessarily lead to an emotion (i.e., running on the treadmill doesn't lead to fear)

(b) Cannon-Bard theory

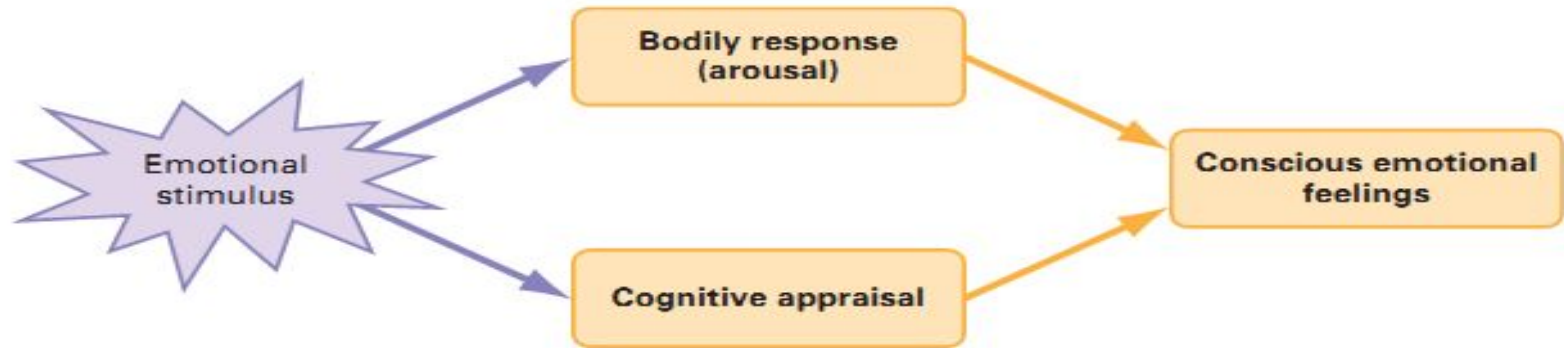
- Their proposal:

- Stimuli **simultaneously** evoke bodily arousal and conscious emotional feelings



Two-factor theory (1962)

- **Physiological arousal** and the **context** in which it's experienced are combined to generate conscious feelings
- i.e., emotions are based on our cognitive appraisal of a situation



Evidence for two-factor theory

- Schacter and Singer (1962)
 - Injected subjects with *epinephrine* (adrenaline)
 - Subjects placed in a room with a *confederate* (an actor) instructed to act either joyful or angry
 - Subjects reported experiencing the same emotion as the confederate!
 - Conclusion: **the same bodily response can be labeled differently depending on the context**

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Manipulating emotion

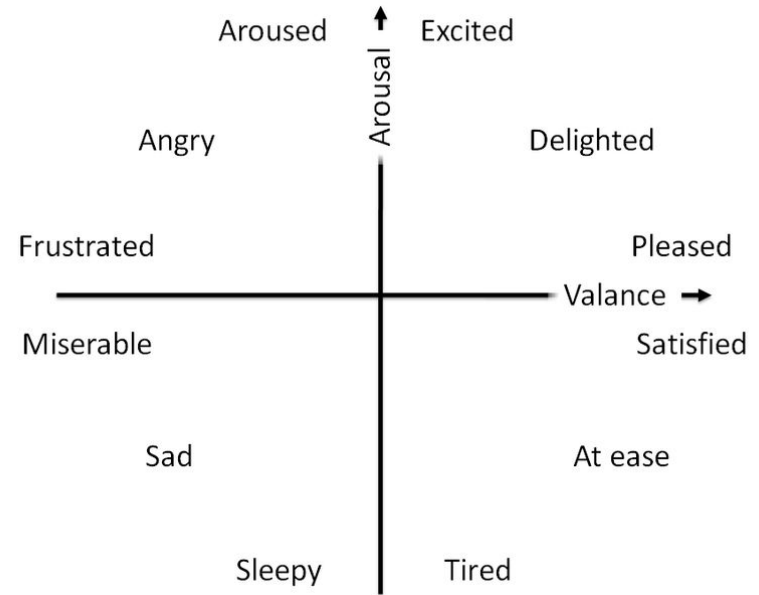
- Rather than rely on subjects' moods upon entering the lab, it's more useful to *induce* a particular mood
 - Present them with emotional stimuli
 - E.g., film clips, music, words, pictures, pictures of facial expressions, etc
 - Give them rewards or punishments
 - E.g., money vs electrical shock

Direct measures of emotion

- Emotions are most commonly assessed via **self-report** measures, i.e., asking subjects to describe or rate their own emotions
- Pros:
 - Direct assessment
- Cons:
 - Relies on introspection
 - Affected by cultural conventions

Direct measures of emotion

- Subjects are often asked to rate their emotional responses along two dimensions...
- **Arousal**: degree of bodily changes that occur during an emotional response
- **Valence**: subjective quality (positive or negative) of an emotional response
- Examples
 - Mildly happy = low arousal, positive valence
 - Extremely scared = high arousal, negative valence



Indirect measures of emotion

- Emotions can also be measured *indirectly* by observing behavior (e.g., a rat freezing) or taking physiological measurements
 - Particularly useful when studying animals
- Popular physiological measure in humans is the **skin conductance response** (SCR)
 - Change in electrical conductivity of the skin dependent upon perspiration level
 - Often used in “lie detector” tests

Outline

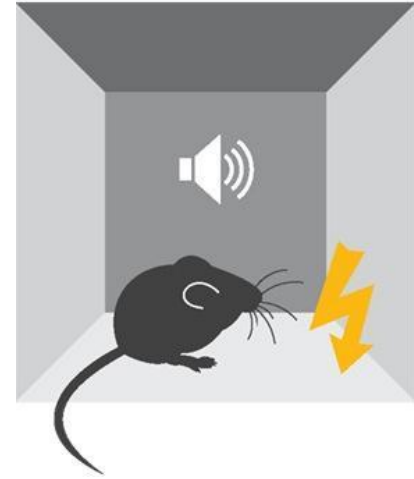
- Defining emotion
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Emotion and conditioning

- Emotional conditioning involves direct experience with emotional stimuli
- **Classical conditioning**
 - Learn about associations between stimuli (US and CS)
 - The US has emotional significance: appetitive (e.g., food) or aversive (e.g., shock)
- **Operant conditioning**
 - Learn about associations between behaviors and their consequences: $R \rightarrow O$
 - The *outcome* has emotional significance: reinforcing or punishing

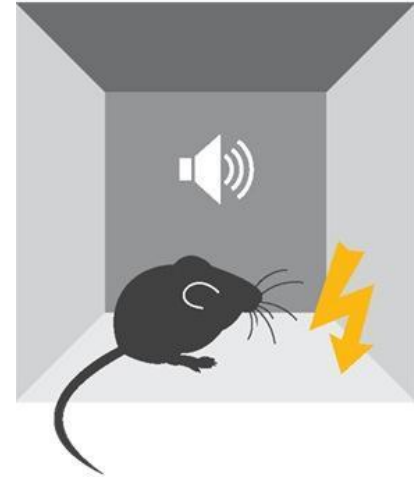
Fear conditioning

- With repeated pairing of Tone (CS) and Shock (US), animals show a fear response (CR) to the tone alone
 - Behavior: freezing
 - Accompanying physiological response: e.g., increased heart rate
- Important in that it allows us to learn to predict danger
- What kind of conditioning is this (classical or operant)?
 - *Classical!*



Fear conditioning

- Fear conditioning can be *very* hard to extinguish!
 - And easy to restore
 - Implies a role for long-term memory
- PTSD likely involves fear conditioning



Phobias

- Phobias often arise through fear conditioning

Sight of dog (CS)



Bite (US)



Fear (CR)



Phobias

- **Phobias** often arise through fear conditioning
 - *phobia*: persistent fear that is typically disproportional to the actual danger posed
- Common therapeutic approach is **exposure therapy**
 - Very similar to the extinction phase of classical conditioning!
 - E.g., incremental exposure to dogs (CS) *without* an aversive event (being bitten; US)

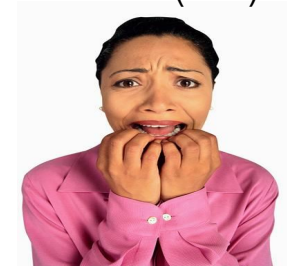
Sight of dog (CS)



Bite (US)



Fear (CR)



Conditioned Escape

- Learning to 'get away' from danger by performing some behavior
- What kind of conditioning is this (operant or classical)?
 - *Operant!*

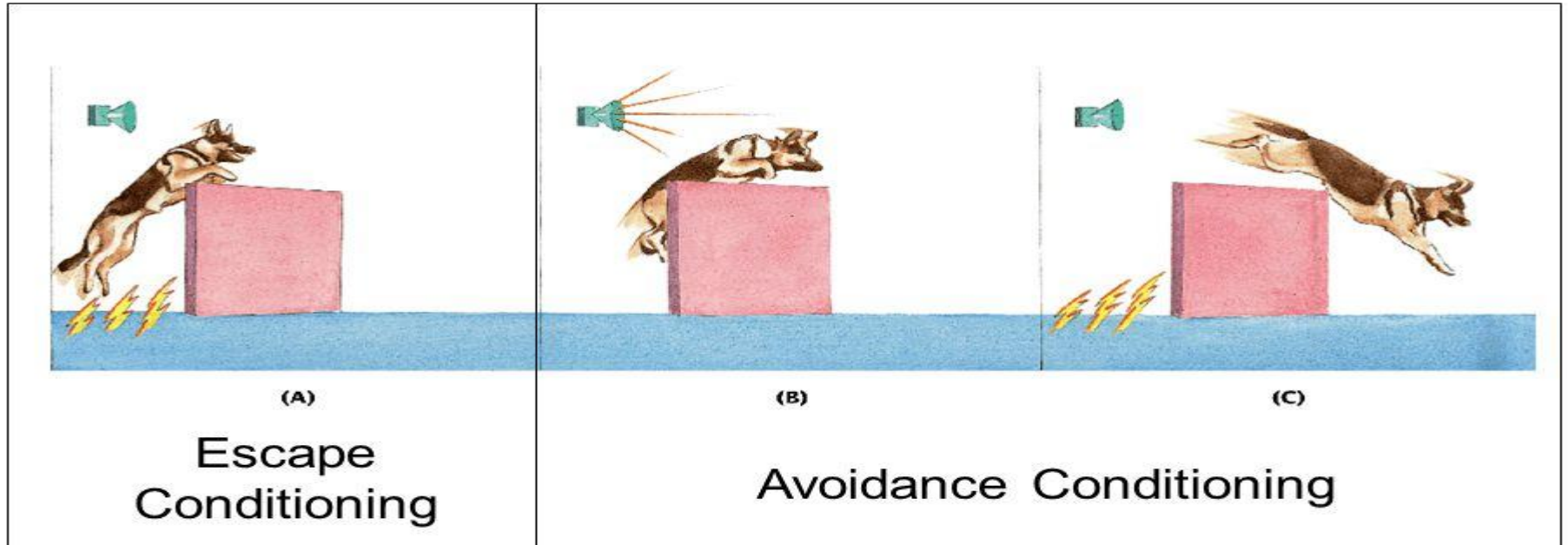
Example: rat learns to 'escape' the water by finding a platform to stand on



Conditioned Avoidance

- Learning to make responses to avoid or prevent an aversive stimulus
- E.g., rat learns that when it hears a tone, if it presses lever, it can *avoid* a foot shock
- What kind of conditioning is this (operant or classical)?
 - *Not quite clear!* Probably a combination of operant and classical conditioning
 - E.g., the rat likely learns a fear response to the tone (**classical conditioning**) but then learns a *behavior* to avoid the fear response (**operant conditioning**)

Escape vs. Avoidance



Adapted from: *The Psychology of Memory and Learning* by Hintzman. © 1978 by W.H. Freeman and Company. Used with permission.

Learned Helplessness

Exposure to an uncontrollable punishment leads to *learning* that behavior (responses) are ineffectual. This reduces/eliminates avoidance responses

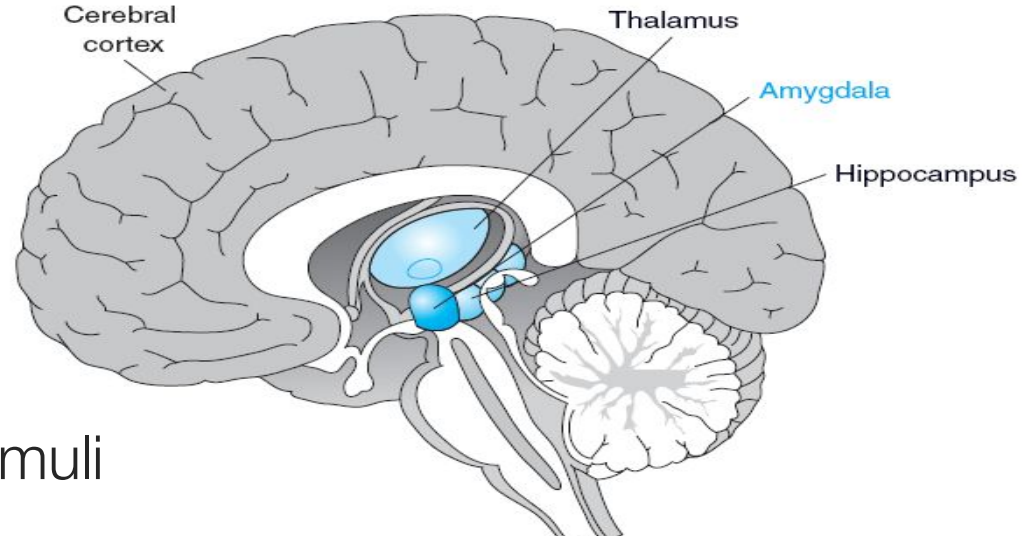


Avoidance task: Dog gets shocked in one chamber. Dog learns to *escape* the shock by jumping over a wall to the safe side. If the shock is preceded by a tone, the dog will *avoid* the shock by jumping over the wall when it hears the tone

However, if the avoidance task is preceded by a classical conditioning task where the tone predicts a shock (and the dog can't jump to a new chamber), then when the dog is in the avoidance task, **it doesn't even try to escape/avoid the shock**. It just lays in the corner and whines.

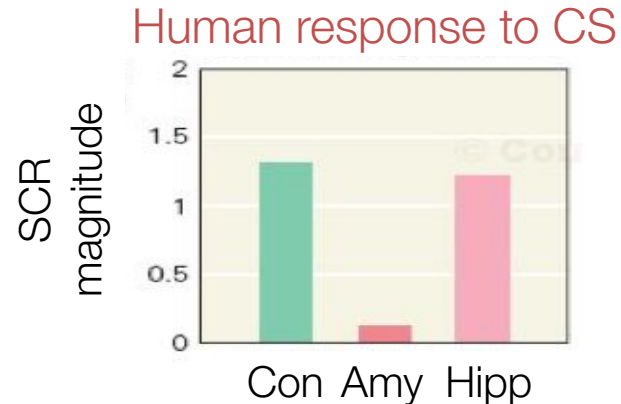
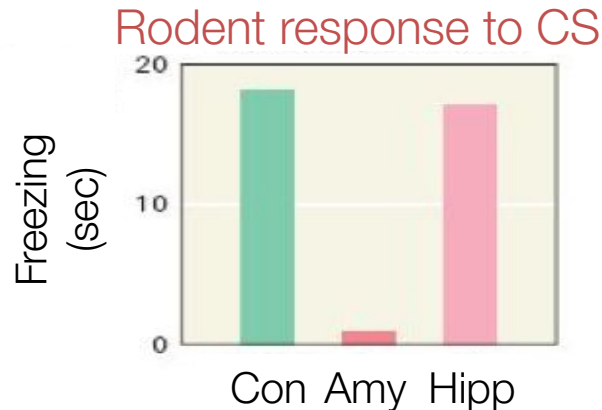
Neural basis of fear conditioning

- **Amygdala:** Greek word for almond
- Sits anterior to the hippocampus
- Plays a key role in:
 - Processing emotional stimuli
 - Expressing emotions
 - Fear learning



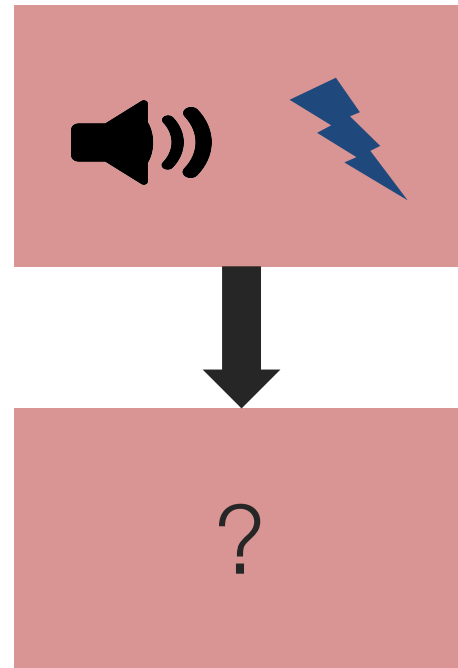
Fear learning

- In rodents and humans, **lesions of the amygdala abolish fear learning**
 - I.e., despite repeated tone (CS) – shock (US) pairings, no fear response to the tone
- However, **lesions of the hippocampus leave fear responses intact**



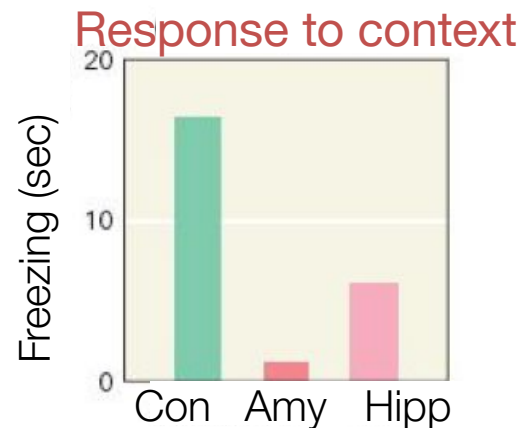
Contextual fear learning

- After repeated CS–US pairings in a particular **context** (e.g., a room), animals show:
 - A fear response to the CS alone
 - A fear response **to the context alone**
- Given the role of the hippocampus in associating an event with its context...
 - Do hippocampal lesions impair fear responses to the context?



Contextual fear learning

- After repeated CS–US pairings in a particular context (e.g., a room), animals show:
 - A fear response to the CS alone
 - A fear response to the context alone
- Given the role of the hippocampus in associating an event with its context...
 - Do hippocampal lesions impair fear responses to the context? **Yes!**



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Memory for emotional events

- Research has examined the following aspects of memory for emotional vs. neutral events
 - Accuracy
 - Vividness
 - Confidence

Cahill and McGaugh (1995)

- Subjects saw a series of pictures accompanied by either an emotional story or a neutral story
- Beginning and end of each story were the same across groups (neutral)
- Middle differed between groups (neutral vs. emotional)
- How well are details remembered for the middle of the story? (memory tested 2 weeks later in surprise test)

Cahill and McGaugh (1995)

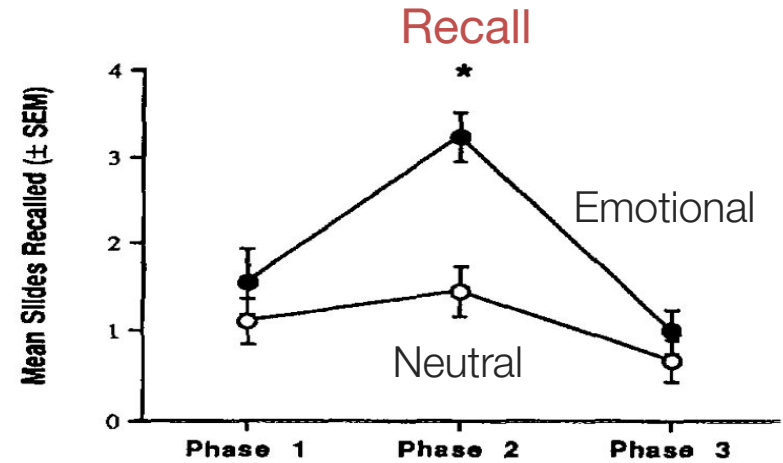
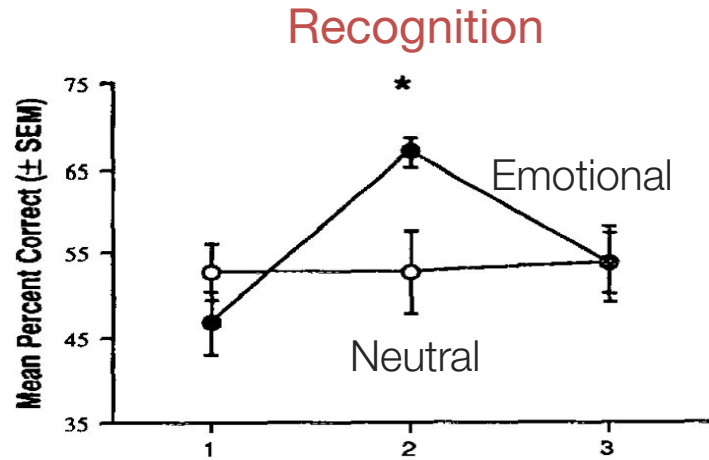
Example of emotional version:

- A boy and his mother are on an outing, and the boy is hit by a car. Surgeons struggle to save his life, and are able to re-attach his severed feet. Later, the father stays with the son while the mother picks up their other child

Example of neutral version:

- A boy and his mother are on an outing, and they pass by the scene of an accident. The boy and his mother visit the hospital, where his father works, and where a disaster-preparedness drill is going on involving life-like surgeries. Later, the father stays with the son while the mother picks up their other child

Cahill and McGaugh (1995)



Why does emotion enhance memory?

- Greater attentional engagement
- Increased physiological arousal
- Increased amygdala activity

Attentional engagement

- **Affective primacy hypothesis**

- Emotional stimuli capture our attention via bottom-up mechanisms
- Less attention goes to non-emotional stimuli

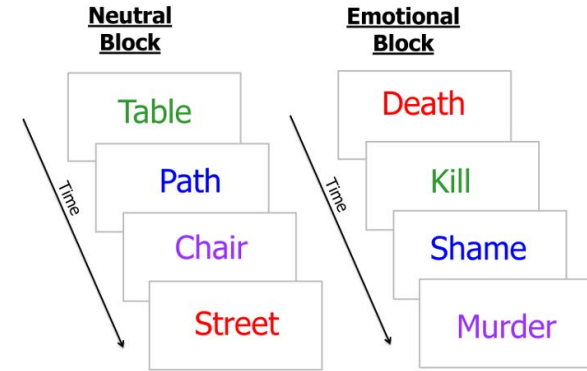
- Evidence

- **Emotional Stroop task**

- More difficult to name ink color of emotional than neutral words

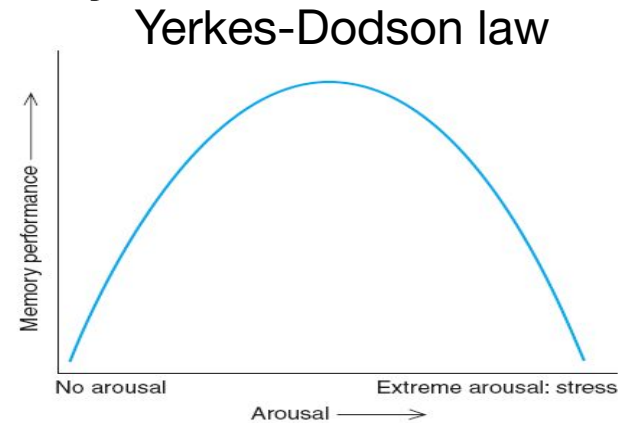
- **Weapons focus task**

- Attention is drawn to the weapon (harming memory for the perpetrator and other details)



Physiological arousal

- Physiological arousal stimulates release of stress hormones like cortisol
- Cortisol increases hippocampal activity
- **Yerkes-Dodson law**
 - **Mild to moderate** arousal leads to **improved** memory performance
 - **Strong** levels of arousal can **inhibit memory** (toxic for the hippocampus)

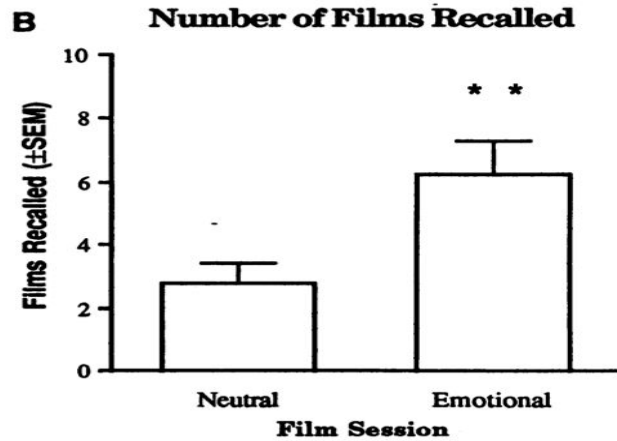


Amygdala activity

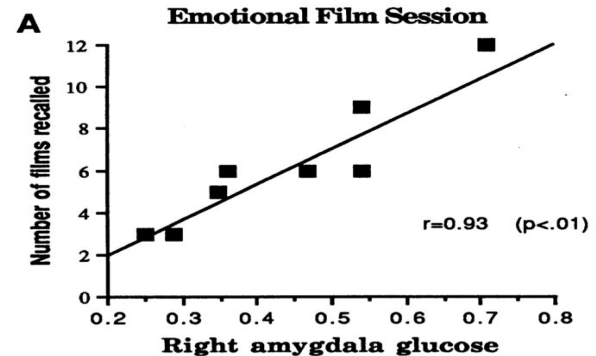
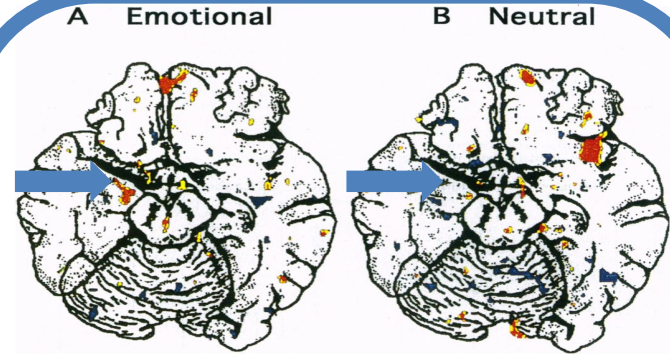
- Cahill et al. (1996)
 - Subjects watched both neutral and negative film clips while in a PET scanner
 - Free recall test three weeks later
- Two questions:
 - Is recall better for emotional than neutral clips?
 - If so, does activity in the amygdala relate to better recall for emotional events?

Cahill et al. (1996)

Behavioral data



Neural data



Memory for emotional events

- Research has examined the following aspects of memory for emotional vs. neutral events
 - Accuracy ✓
 - Vividness
 - Confidence
- Higher ratings for emotional than neutral stimuli, including flashbulb memories

Conflict?

- Many studies demonstrate that memory is more accurate for emotional than neutral events
- Other evidence shows that flashbulb memories are no more accurate than everyday memories
- Possible explanation?
 - Emotion may not enhance memory equally for all aspects of an emotional event
 - Emotion tends to **enhance memory for *central* aspects** of an event, but may **impair memory for peripheral aspects**
 - *tradeoff* of divided/narrowed attention

Kensinger et al. (2007)

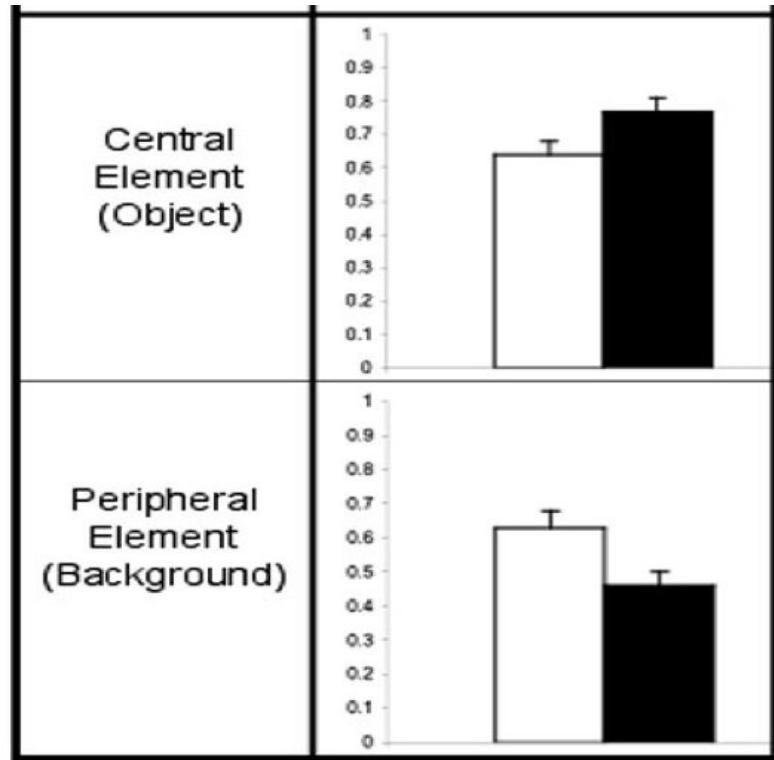
- Subjects studied a series of scenes with either neutral or negative objects inserted
- Recognition test 30 minutes later for both the central objects and the background scenes

STUDY PHASE:



Approach or move away?

Kensinger et al. (2007)



□ Neutral object
■ Negative object

Suggests that emotion induces a *tradeoff*

Negative, central elements benefit at the expense of neutral background elements

Emotion-based biases

Emotion doesn't only influence memory at encoding – the mood you are in at *retrieval* can influence the content you recall

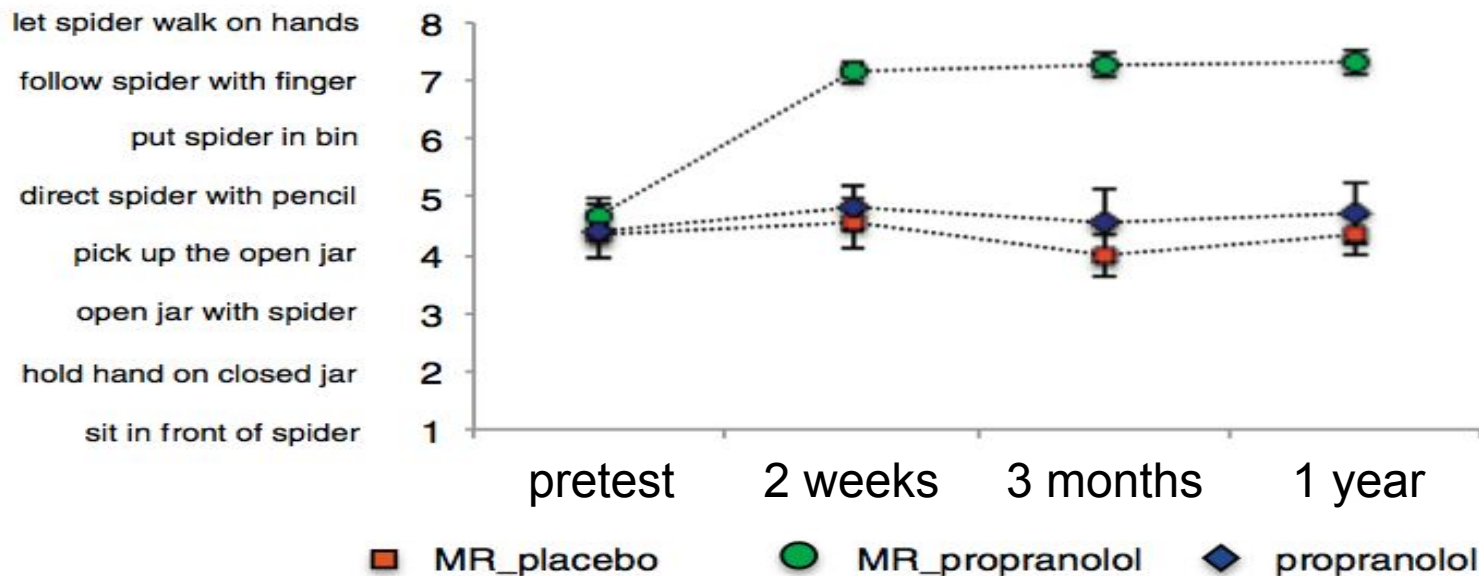
- E.g., being in a sad mood can bias you toward recalling negative events
- **Mood congruent memory**
- This is relevant (and problematic) in depression

Propranolol

- Beta-blocker drug originally used to treat hypertension
- Blocks the action of norepinephrine (a neurotransmitter that triggers adrenergic activity)
- Adrenergic activity may be essential for the enhancement of emotional memories
- People who show greater signs of adrenergic activity (e.g. racing heart rate, panic behavior) immediately after a traumatic event are more likely to exhibit PTSD symptoms later
 - *evidence for arousal as critical to emotional memory consolidation?*
- Being researched as a method of preventing or treating PTSD and other anxiety disorders

Propranolol During Reconsolidation

Individuals with arachnophobia were exposed to a tarantula for two minutes [memory reactivation (MR)], then given propranolol.



- Effective at targeting older, strong memories!

Next Up:

- **Quiz #3!**